

Carrier: An Experimental Platform for Multimodal Mixed-Agent Social Simulation

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ABSTRACT

Large Language Models (LLMs) increasingly demonstrate stable role enactment, socio-emotional attunement, and coordinated behavior in multi-agent contexts, suggesting that they can function as actors within collective social systems rather than isolated text-generation tools. These capacities are highly relevant to policymaking, where decisions are collective, high-stakes, and seldom permit controlled field experimentation. Mixed-agent simulation—combining humans, LLM agents, and scripted actors—provides an *in silico* paradigm for studying deliberation, coordination, and information diffusion under controlled variation. Yet existing systems are often bespoke and weakly instrumented, limiting construct validity and comparability. We introduce Carrier (<https://www.carrierlab.org>), an open-source platform for multimodal mixed-agent simulation. Carrier structures interaction into modular episodes and defined roles, enabling systematic manipulation of agency, authority, and disclosure. With standardized instrumentation and process tracing, it supports mechanism-focused analysis of coordination, delegation, responsibility attribution, and collective error detection in simulated policy contexts.

CCS CONCEPTS

• Human computer interaction (HCI) • Interaction paradigms • Web-based interaction

KEYWORDS

Mixed-agent simulation; Collective behavior; Human–AI collaboration; Experimental methodology; Multimodal systems

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1 Introduction

Large Language Models (LLMs) are transitioning from probabilistic text generators to agentic systems capable of sustained, role-constrained social interaction. Experimental studies indicate that LLMs can reliably instantiate instructed interpersonal styles, demonstrating controllable variation in role enactment [1,7]. In applied dialogue settings, their responses meet counselling-relevant process criteria—such as empathic reflection and adherence to motivational interviewing principles—suggesting competence in structured socio-emotional support [2–4]. Beyond dyadic interaction, memory-augmented generative agents exhibit temporally coherent behavior, relationship persistence, and emergent collective dynamics, including coordination and information diffusion in simulated social environments [5,6]. Taken together, this evidence suggests that LLMs can approximate stable role behavior, socio-emotional attunement, and foundational forms of collective social organization under controlled conditions.

These developments are consequential for policymaking, where decisions are collective, high-stakes, and marked by deep uncertainty, yet seldom permit randomized field experimentation. Ethical constraints, legitimacy concerns, time pressure, and political cost limit real-world manipulation, and counterfactuals are rarely observable. Mixed-agent simulation—combining humans, LLM agents, and scripted actors—provides a complementary *in silico* paradigm for studying deliberation, coordination, polarization, and information diffusion under controlled variation. By manipulating group composition, incentive structures, disclosure regimes, and interaction modalities, such environments allow policymakers to stress-test communication strategies, model social influence dynamics, and assess robustness across alternative behavioral assumptions.

However, current approaches fall short of the standards required for psychologically grounded policy simulation. Many systems are bespoke and text-centric, with limited

experimental control over role operationalization, incentive structures, and memory dynamics. This constrains construct validity and replication, and risks overinterpretation driven by face validity rather than systematic evidence.

To address these limitations, we introduce Carrier (<https://www.carrierlab.org>), an open-source, browser-based platform for multimodal mixed-agent simulation and controlled behavioral experimentation. Carrier supports reusable interaction protocols, explicit role assignment across human and artificial agents, and systematic variation in communication modality while preserving experimental structure. It provides standardized instrumentation, embedded behavioral measures, and fine-grained event logging—including agent memory states and input-output traces—to enhance transparency, auditability, and construct validation. By lowering engineering barriers and embedding methodological rigor, Carrier enables participatory, psychologically informed policy simulation that is both experimentally tractable and epistemically cautious. This offers a highly controllable human-agent simulation environment focusing on smaller group interactions, providing a natural companion to existing platforms Sotopia [8] or CAMEL [9], which primarily offer large scale simulation with an emphasis on agent performance and interaction.

2 Carrier: Infrastructure for mixed-agent social simulation

Carrier is designed as experimental infrastructure for mixed-agent social simulation, in which multiple human participants interact with artificial agents within a shared interaction episode. Rather than conceptualizing automation as a single treatment, Carrier operationalizes the distribution of agency as an experimentally manipulable variable—specifying who initiates behavior, who holds decision authority, what information is visible, how communication flows, and how participants can intervene, override, or correct system outputs.

This architecture enables factorial manipulation of group composition and role structure (e.g., dyads vs. triads; single- vs. multi-agent collectives; disclosed vs. undisclosed agent presence) while holding procedural context constant. Methodologically, Carrier supports controlled variation in interaction structure, allowing researchers to isolate causal mechanisms underlying coordination, social influence, delegation, and responsibility attribution within collective decision-making contexts.

2.1 Role architecture: type × role

Carrier formalizes mixed-agent interaction using a Type × Role schema. Type specifies the ontological status of an actor (human participant or agent), while Role specifies the functional locus at which agency enters the workflow. This separation reduces role ambiguity and makes “who did what”

analytically tractable. Carrier currently implements three primary roles:

Communicator. Communicators participate directly in the shared interaction stream. Human and artificial communicators occupy the same interface, preserving interactional symmetry and enabling controlled comparisons across human-only and mixed-agent compositions. Communicator behavior can be constrained through protocol rules (e.g., turn-taking, timed phases, structured deliberation), supporting experimental manipulation of discourse norms and influence processes.

Mediator. Mediators operate at the coordination layer rather than as peer contributors. They monitor interaction dynamics and deliver structured interventions (e.g., prompts, summarizations, attention cues, rule enforcement) triggered by temporal schedules or behavioral thresholds. This enables systematic study of facilitation strategies, governance mechanisms, and procedural justice within deliberative settings.

Processor. Processors function at the drafting layer, shaping content before public commitment. They assist via review, rewriting, or generation while preserving human veto authority. This role enables examination of back-stage cognitive offloading, delegation, reliance, and disclosure effects—particularly how unseen algorithmic influence alters attribution, trust calibration, and perceived authorship.

By holding capability constant while varying role placement, Carrier supports causal tests of how identical computational resources produce divergent collective outcomes depending on their position within the social ecology.

2.2 Experimental flow: chambers and run plans

Carrier decomposes a study into chambers—atomic, synchronous interaction episodes with a defined composition (human/agent/bot slots), roles, modality, and optional pre/post measures. Chambers are then sequenced into condition paths (chamber lines), allowing researchers to vary collaboration structure across steps while keeping the overall protocol and measurement scaffold comparable across conditions.

Group formation is handled by a structured matching layer that operationalises eligibility constraints, timeouts, and fallbacks into repeatable assignment logic. Participant measures collected via surveys (e.g., domain expertise, trust propensity, risk orientation) can be used as matching variables and as parameters for agent behaviour (e.g., stance, assertiveness, supportiveness), enabling simulations and studies with heterogeneous populations. Crucially, Carrier records the full assignment provenance—who was eligible, what rule fired, which fallback was used, and the resulting exposure sequence—supporting reproducibility and post hoc causal tracing.

This architecture preserves key ecological properties of real collectives—staggered arrivals, uneven participation, dropout, and mixed compositions—while maintaining experimental control over who interacts with whom, under what roles, and in what order.

2.3 Instrumentation and process tracing

A central design principle of Carrier is instrumented transparency. The platform logs interaction events, role states, agent prompts, memory states, and outputs within a standardized schema. This supports process tracing of the full causal chain from initial configuration to downstream collective outcomes.

Carrier captures fine-grained behavioral measures relevant to human–AI collaboration research: draft histories, suggestion acceptance/rejection, time-on-task, revision and undo patterns, frequency of agent invocation, turn-taking dynamics, and latency metrics. These measures enable analysis of delegation, oversight, reliance, and out-of-the-loop effects using established behavioral science frameworks.

Operationally, Carrier integrates study specification, participant runtime, and experimenter oversight within a unified system. Experimenter interventions (e.g., pause, skip, termination) are logged to preserve an auditable record of protocol deviations, enhancing internal validity and replication fidelity.

3 Conclusions, limitations and future work

Large Language Models (LLMs) are rapidly evolving from text generators into reasoning systems capable of acting as autonomous agents. When embedded in social contexts, these agents can exhibit emergent behaviors such as coalition formation, information diffusion, norm propagation, and collective decision-making. Carrier provides the infrastructure needed to study these dynamics under controlled conditions. By structuring mixed-agent interaction into configurable roles and modular interaction episodes, Carrier allows researchers to systematically vary agency distribution, decision rights, visibility, and intervention pathways within heterogeneous groups. Standardized instrumentation and process tracing make collective dynamics analytically tractable, enabling researchers and policymakers to examine how communication strategies, incentives, and disclosure regimes shape coordination, influence cascades, and unintended consequences before real-world deployment.

Despite these contributions, limitations remain. Real-time AI-driven video agents are not yet supported, constraining certain multimodal designs. Web-based deployment introduces latency variability that can affect fine-grained temporal measures [10], and current WebRTC infrastructure places practical limits on group size and session quality. Moreover, while Carrier standardises experimental scaffolding,

rigorous mixed-agent studies still require careful specification of behavioural assumptions and incentive structures to maintain construct validity.

Future work will strengthen methodological robustness and scalability. Planned extensions include expanded controls over agent memory and role configuration, improved support for heterogeneous populations, enhanced multimodal stability, and reference paradigms for cross-study benchmarking. We also aim to formalise participatory calibration workflows that allow domain experts to iteratively refine assumptions and stress-test protocol designs. Through these developments, Carrier advances mixed-agent simulation as a reproducible, experimentally tractable method for studying collective decision-making in complex socio-technical systems.

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